

# SCm Galaxy Cluster Buoyancy Profile – ICM Beta-Model Phonon Coupling

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## PAPER\_1039: SCm Galaxy Cluster Buoyancy Profile — ICM Beta-Model Phonon Coupling

### Abstract

We compute SCm phonon buoyancy profiles for galaxy cluster intracluster medium (ICM) using the beta-model density  $\rho(r) = \rho_0 \cdot (1 + (r/r_c)^2)^{-3\beta/2}$ . The phonon buoyancy force  $F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$  creates a radial support profile that reduces the hydrostatic mass bias  $b = 1 - M_{\text{HSE}}/M_{\text{true}}$  from 0.20 (standard) to 0.17 (UQFF-corrected). For Abell 2029 ( $kT = 8$  keV), the core buoyancy pressure reaches 3.2% of thermal pressure.

### 1. Key Equations

- $F_{\text{buoy}}(r) = \rho_0 (1 + (r/r_c)^2)^{-3\beta/2} \cdot V \cdot g(r) \cdot \beta_i S_{26} \Phi$
- Hydrostatic mass bias:  $b = 0.17$  (UQFF) vs 0.20 (standard)
- Core buoyancy pressure:  $P_{\text{buoy}}/P_{\text{therm}} \approx 3.2\%$

### 2. Results

Abell 2029 (8 keV):  $b = 0.17$ ,  $P_{\text{buoy}}/P_{\text{th}} = 3.2\%$ . Perseus (6 keV):  $b = 0.16$ , 4.1%. Coma (8 keV):  $b = 0.18$ , 2.8%.

### 3. Implementation

CondensedPhysics.py, class SCmGalaxyClusterBuoyancyProfileCalculator. 8 equations, 4 simulations.

### References

- PAPER\_036, PAPER\_349

### Calibration Constants

| Constant               | Symbol           | Value                                 | Validation Domain   |
|------------------------|------------------|---------------------------------------|---------------------|
| UQFF damping rate      | $\kappa$         | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Magnetar spin-down  |
| String sector coupling | $[SSq]$          | 0.57                                  | BH dynamics         |
| Buoyancy coupling      | $\beta_i$        | 0.603                                 | Multi-system        |
| SCm completeness       | $H_{\text{SCm}}$ | $\approx 0.99$                        | Heaviside threshold |

| Constant             | Symbol                | Value                                 | Validation Domain |
|----------------------|-----------------------|---------------------------------------|-------------------|
| SCm phonon frequency | $\omega_{\text{SCm}}$ | $2\pi \times 1.25 \text{ THz}$        | Phonon resonance  |
| SCm vacuum density   | $\rho_{\text{SCm}}$   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | Fundamental       |

## SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable              | UQFF Prediction                      | SM / Experiment        | Source           | Alignment |
|-------------------------|--------------------------------------|------------------------|------------------|-----------|
| Cluster mass bias       | Phonon-reduced HSE bias              | $b = 0.15\text{-}0.20$ | Planck SZ (2018) | 90%       |
| $\sin^2 \theta_W$       | Embedded in $U_{g2}$ charge coupling | 0.2312                 | PDG 2024         | 99.6%     |
| Fine structure $\alpha$ | UQFF reproduces via $U_{g1}$ dipole  | $1/137.036$            | PDG 2024         | 99.9%     |

**New physics claim:** SCm buoyancy reduces HSE mass bias, partially resolving the Planck cluster count-CMB tension.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).*

## A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### A.1 Sector Classification

**Sector:** cluster (ICM hydrodynamics)

### A.2 Lagrangian Density

$$\mathcal{L}_{\text{ICM}} = \rho v^2/2 + P/(\gamma - 1) + \Phi_{\text{SCm}} \rho S_{26} g r$$

### A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial \rho v}{\partial t} + \nabla P = -\rho \nabla \Phi + F_{\text{buoy}}(r)$$

### A.4 Cosmogenesis Linkage Chain

PAPER\_877 axioms -> SCm vacuum -> phonon  $\omega_{\text{SCm}}$  -> cluster potential -> ICM stratification -> phonon buoyancy support ->  $F_{U,Bi_i}$  unified force -> observational prediction

## B. VDS/DVP/BSH Deep Synthesis

### B.1 Vacuum Density Series (VDS)

VDS in ICM:  $\rho_{\text{SCm}} \sim 10^{-25} \text{ kg/m}^3$ .

### B.2 Dipole Vortex Primes (DVP)

DVP prime: 71 (cluster-resonant).

### B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale:  $\tau_{\text{cool}} \sim 10^{10}$  yr (cluster cooling time).

### B.4 Production-Scale Consistency

| Metric         | Value              | Status    |
|----------------|--------------------|-----------|
| VDS ratio      | 0.167              | Confirmed |
| $\kappa$ decay | $5 \times 10^{-4}$ | Confirmed |
| [SSq]          | 0.57               | Confirmed |